

Lecture 1 - 8/26

What is number theory?

Number theory is a broad area. Here is a succinct description (and over-simplification) of it. Number theory is the study of solutions to equations (or systems of equations) where the values of the variables must be whole numbers (integers) or ratios of whole numbers (rational numbers). Despite the simplicity of the problems, there is still a huge amount we do not know about these things.

Examples:

1. A prime number n is a number divisible only by itself and 1. Another way of saying this is that a prime number is one for which the equation $n = xy$ has exactly four solutions with x and y integers $((n, 1), (1, n), (-n, -1), (-1, -n))$.
2. The equation $x^3 + y^3 = 1$. The points $(1, 0)$ and $(0, 1)$ are solutions. It is not hard to prove that these two points are the only points with integer coordinates on the curve. It is much harder to prove that $(1, 0)$ and $(0, 1)$ are the only points on the curve where x and y are *rational*. (This was done by Leonhard Euler in 1753.)
3. Integer solutions to $x^2 + y^3 = z^7$. There are many integer solutions, but if we impose the condition that no two of x , y and z have a common factor, there are exactly 16 solutions, namely

$$(\pm 1, -1, 0), (\pm 1, 0, 1), \pm(0, 1, 1), (\pm 3, -2, 1), (\pm 71, -17, 2) \\ (\pm 2213459, 1414, 65), (\pm 15312283, 9262, 113), (\pm 21063928, -76271, 17).$$

This was proven in 2007.

4. Integer solutions to $x^3 + y^3 + z^3 = 3$.

One obvious solution is $x = y = z = 1$. One slightly less obvious solution is $x = 4$, $y = 4$, $z = -5$.

In 1953, Mordell wrote “it must be very difficult indeed to find out anything about any other solutions.” In 2019, Booker and Sutherland found a solution with

$$x = 569936821221962380720 \\ y = -569936821113563493509 \\ z = -472715493453327032.$$

The difference between MTH 345/645 and MTH 346/646

Wake Forest offers two number theory courses: Elementary Number Theory (MTH 345/645), and Modern Number Theory (MTH 346/646). The former is taught every spring, and the latter is taught once every two years (in the fall).

Elementary number theory covers material that was known to the Greeks and developed by European mathematicians before 1800. The objects are very basic, and in this course it is common to reason from first principles. However, this course (more so than perhaps most other undergrad classes) that it can be very difficult to see the path from the problem to the solution. I usually teach this course using “inquiry-based learning” - the students follow a textbook (which has no proofs) and supply the proofs to build up the subject.

Modern number theory is not always taught on the same topic, but the topic of elliptic curves really became a thing in the early 20th century, and is at the forefront of current research. MTH 346/646 will end with a discussion of an unsolved problem, a proof of which is worth 1 million dollars.

This course will have more theory that is used in it, but it will be much less common to reason from first principles. It will be more common to use the theoretical knowledge we have to approach problems, and this makes it easier to sit down and have an idea about how to start a problem.

What is an elliptic curve?

An elliptic curve is an equation of the form $y^2 = x^3 + ax^2 + bx + c$ (where a , b and c are usually taken to be integers), although slightly more general forms of this equation are OK too.

(Draw a picture.)

Elliptic curves are not ellipses. The theory of elliptic curves begins with Legendre’s work in 1811 on elliptic integrals and elliptic functions. This theory arose out of trying to find the arc length of an ellipse. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (with $a \geq b$) can be parametrized by $x = a \cos(t)$, $y = b \sin(t)$, and after a few trig identities and the arc length formula, we get that the arc length is

$$\int_0^{\pi/2} \sqrt{1 - k^2 \sin^2(\theta)} d\theta$$

where $k = \sqrt{1 - (b/a)^2}$. The problem is that this integral is not easy to do! There is no nice antiderivative that can be found for $\sqrt{1 - k^2 \sin^2(\theta)}$ for $0 < k < 1$. Some sort of numerical approximation for this is necessary. It’s possible to make a change of variable to turn this into an integral of the sort

$$\int_{\lambda}^{\infty} \frac{1}{\sqrt{x(x-1)(x-\lambda)}} dx,$$

and this becomes connected with $E : y^2 = x(x-1)(x-\lambda)$.

Why are they cool? What are we going to learn about them this semester?

1. If E is an elliptic curve, there is a method of “adding points” on E to get new points from old ones. If the starting points P and Q are rational (that is, both x and y coordinates of P and Q are rational numbers), then the new point $P + Q$ is also rational. Sometimes, an elliptic curve can have infinitely many rational points on it. One of the main focuses this semester will

be on how to figure out whether there are only finitely many or infinitely many such solutions. This operation makes the set of points an elliptic curve have the structure of a group. (More on what that means later.)

2. There are a huge number of number theoretic questions that boil down to the problem of studying integer solutions or rational solutions to points on elliptic curves. Here are some examples.

(a) PDF presentation about “Facebook fruit meme.”

(b) A perfect square is a number of the form m^2 . A pyramidal number is a number of the form $1^2 + 2^2 + \cdots + n^2$. (Because you can arrange spheres or cubes in the shape of a pyramid.)

Which numbers are both square and pyramidal?

Squares: 1, 4, 9, 16, 25, 36, 49, 64, ...

Pyramidal: 1, 5, 14, 30, 55, 91,

To be square and pyramidal, we need $m^2 = 1^2 + 2^2 + \cdots + n^2$. Using the formula $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$, we get $m^2 = \frac{n(n+1)(2n+1)}{6}$. This is really an elliptic curve. To make it look like one, multiply by $5184 = 72^2$ (which is a square, and is also three times a cube). We get

$$5184m^2 = 1728n^3 + 2592n^2 + 864n = (12n)(12n + 6)(12n + 12).$$

Now, set $X = 12n + 6$ and $Y = 72m$. We get $Y^2 = (X - 6)X(X + 6) = X^3 - 36X$.

This is an elliptic curve. It turns out that there are infinitely many rational points on it, but the only integers points on it. (It's a theorem that an elliptic curve only has finitely many integer points.)

The only integers points on $Y^2 = X^3 - 36X$ are those with $X = -6, -3, -2, 0, 6, 12, 18$ and 294 . To get back to a non-negative value of n , we need X to be 6 more than a multiple of 12. This happens for $X = 6$ (and $n = 0$), $X = 18$ (and $n = 1$ - the number 1^2 is both square and pyramidal), and $X = 294$ (and $n = 24$). This gives us the solution

$$1^2 + 2^2 + 3^2 + \cdots + 24^2 = 70^2.$$

(c) Fix a positive integer n . Is there a right triangle where all three sides (the two legs and the hypotenuse) have rational length, and where the area of the triangle is n ? (We say that n is a congruent number if there is such a triangle.)

We're looking at rational solutions to the system of equations

$$\begin{aligned} a^2 + b^2 &= c^2 \\ \frac{1}{2}ab &= n. \end{aligned}$$

The 3,4,5 right triangle shows that $\frac{1}{2}(3 \cdot 4) = 6$ is congruent. The 5,12,13 right triangle shows that $\frac{1}{2}(5 \cdot 12) = 30$ is congruent. The 7,24,25 right triangle shows that $\frac{1}{2}(7 \cdot 24) = 84$ is congruent. Dividing the side lengths by 2 gives 7/2, 12, 25/2 shows that 21 is congruent.

It turns out that solutions to this system of equations are in one-to-one correspondence with points on $y^2 = x^3 - n^2x$ with $y \neq 0$.

Using this, it is possible to prove that 1, 2, 3 and 4 are not congruent, but 5, 6 and 7 are. For example, when $n = 5$, we have a solution with $a = \frac{20}{3}$, $b = \frac{3}{2}$ and $c = \frac{41}{6}$. However, since the elliptic curve $y^2 = x^3 - 25x$ has infinitely many rational points on it, there are more. The next one is

$$a = \frac{1519}{492}, b = \frac{4920}{1519}, c = \frac{3344161}{747348}.$$

3. Elliptic curve cryptography.

From about 2015 through late last year, elliptic curve cryptography was the main form of public-key cryptography used. The basis of this cryptosystem is based on the difficulty of the elliptic curve discrete logarithm problem:

Given a point P on an elliptic curve and a point $Q = mP$ on the curve, find m .

One potential issue with it is that a sufficiently powerful quantum computer could break elliptic curve cryptography (and other related cryptosystems like RSA) very quickly. In August of 2024, NIST released the first standard for a “post-quantum” cryptographic system (Kyber or ML-KEM). Shortly thereafter, some browsers (like Chrome and Safari) and some servers (like Cloudflare) introduced support for a hybrid cryptosystem that combines elliptic curve cryptography with the new supposedly quantum-resistant system.